

NTANDOMODS ICT DIPLOMA

MONTH 3

Microsoft Excel

Spreadsheets, Formulas & Data Management

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Introduction to Microsoft Excel

Microsoft Excel is a spreadsheet programme used to organise data, perform calculations, and create charts. It is used by every business in Zimbabwe for stock records, budgets, payroll, sales tracking, and much more.

If you work in retail, you will use Excel to track stock levels, calculate profits, and create reports. If you start your own business, Excel will be your accountant until you can afford one.

Learning Outcomes

- Understand the Excel interface: cells, rows, columns, sheets
- Enter and format data in a spreadsheet
- Use basic formulas: SUM, AVERAGE, COUNT, MIN, MAX
- Create simple charts and graphs
- Sort and filter data
- Use cell referencing (relative and absolute)
- Print spreadsheets correctly

Unit 1: The Excel Interface

1.1 Opening Excel

1. Click Start menu
2. Type 'Excel' in the search bar

3. Click Microsoft Excel
4. Select 'Blank workbook'

1.2 Understanding the Spreadsheet

A spreadsheet is made of cells arranged in rows and columns.

- Columns run vertically and are labelled with letters: A, B, C...
- Rows run horizontally and are labelled with numbers: 1, 2, 3...
- A cell is the box where a row and column meet. Cell B3 means Column B, Row 3.
- A worksheet (sheet) is one page of cells. A workbook is the entire Excel file.

1.3 Key Parts of the Interface

Element	Purpose
Name Box	Shows the current cell address (e.g. A1)
Formula Bar	Shows what you have typed in the current cell
Column Headers	Letters A-Z at the top
Row Numbers	Numbers 1, 2, 3... on the left side
Sheet Tabs	Bottom of screen - switch between sheets
Ribbon	Toolbar with tabs: Home, Insert, Formulas, Data, etc.

Unit 2: Entering and Formatting Data

2.1 Types of Data

Data Type	Example	Alignment
Text (labels)	Product Name, Date	Left-aligned
Numbers	150, 3.75, -20	Right-aligned
Formulas	=A1+B1, =SUM(A1:A10)	Right-aligned (shows result)
Dates	01/06/2026	Right-aligned

2.2 Formatting Cells

- Bold headers: select the row, press Ctrl + B
- Number format: right-click > Format Cells > Number (set decimal places)
- Currency format: right-click > Format Cells > Currency (adds \$ or ZiG symbol)
- Column width: double-click the column border to auto-fit
- Borders: Home tab > Borders button > All Borders
- Fill colour: Home tab > Fill Color (paint bucket icon)

2.3 Practical Exercise: Create a Stock List

Create the following table in Excel:

Item	Qty In Stock	Unit Price	Total Value	Supplier
Rice 2kg	50	\$3.50	\$175.00	OK Zimbabwe
Cooking Oil 2L	30	\$4.00	\$120.00	TM Pick n Pay
Sugar 2kg	45	\$2.80	\$126.00	Spar
Bread	100	\$1.50	\$150.00	Lobels
Eggs (tray)	20	\$5.00	\$100.00	Irvines

Unit 3: Formulas and Functions

3.1 What is a Formula?

A formula is an instruction that tells Excel to calculate something. Every formula starts with an equals sign (=).

3.2 Basic Operators

Operator	Meaning	Example
+	Add	=A1+B1
-	Subtract	=A1-B1
*	Multiply	=A1*B1

/	Divide	=A1/B1
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3.3 Essential Functions

Function	What it Does	Example
SUM	Adds up a range of cells	=SUM(B2:B10)
AVERAGE	Calculates the mean average	=AVERAGE(B2:B10)
COUNT	Counts how many cells have numbers	=COUNT(B2:B10)
MAX	Finds the highest value	=MAX(B2:B10)
MIN	Finds the lowest value	=MIN(B2:B10)
IF	Tests a condition	=IF(A1>50,"Pass","Fail")

3.4 Cell References

Relative Reference (e.g. A1)

Changes when you copy the formula to another cell. If you copy =A1+B1 from row 1 to row 2, it becomes =A2+B2.

Absolute Reference (e.g. \$A\$1)

Does NOT change when copied. Use the \$ sign to lock the row, column, or both. Press F4 after clicking a cell to toggle.

Unit 4: Charts and Data Management

4.1 Creating a Chart

1. Select the data you want to chart (including headers)
2. Go to Insert tab > Chart
3. Choose the chart type: Column, Bar, Pie, or Line
4. Click OK - the chart appears on your sheet
5. Click on the chart to edit titles, colours, and labels

When to Use Each Chart Type

- Column/Bar chart: comparing quantities (sales per month, stock levels)

- Line chart: showing trends over time (monthly revenue)
- Pie chart: showing parts of a whole (market share, expenses breakdown)

4.2 Sorting Data

1. Click any cell in the column you want to sort by
2. Go to Data tab > Sort A to Z (ascending) or Sort Z to A (descending)
3. For custom sorts: Data tab > Sort > Add Level

4.3 Filtering Data

1. Select your data range including headers
2. Go to Data tab > Filter
3. Click the dropdown arrow in any column header
4. Select or deselect values to filter

Filtering hides rows that do not match your criteria. The data is still there - just hidden temporarily.

4.4 Printing a Spreadsheet

- Set Print Area: select the range > Page Layout > Print Area > Set Print Area
- Preview: File > Print (Ctrl + P)
- Fit to one page: in Print settings, change 'No Scaling' to 'Fit Sheet on One Page'
- Add headers/footers: Page Layout > Print Titles

Month 3 Assessment

Theory (30 marks)

1. What is the difference between a row and a column? (4 marks)
2. Write the formula to add cells A1 to A10. (4 marks)
3. What is the difference between a relative and absolute cell reference? Give an example of each. (6 marks)

4. Name 3 chart types and when you would use each one. (6 marks)
5. What does the AVERAGE function do? Write the formula. (4 marks)
6. Explain what filtering does and how to apply it. (6 marks)

Practical (70 marks)

1. Create a monthly budget spreadsheet with Income and Expenses columns. Use SUM to total each category. (20 marks)
2. Create a sales report with at least 10 products. Include formulas for total revenue, average price, highest and lowest sale. (20 marks)
3. Create a column chart from your sales data showing revenue by product. (15 marks)
4. Sort your sales data from highest to lowest revenue. Apply a filter to show only products over \$50. (15 marks)

Pass Mark: 60%